

Bayesian-based analysis for modal identification via cost-friendly devices

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ABSTRACT: It is crucial to assess the condition of existing civil infrastructure systems continuously to ensure public safety and sustainability. However, in many applications of Structural Health Monitoring (SHM), conventional approaches often rely on high-cost measurement devices, which require complex installation, limiting their feasibility for long term applications. The core of this paper is to investigate the performance of low-cost accelerometer sensors integrated into smartphones for system identification and quantify the uncertainties of modal parameters of a pedestrian bridge. A Bayesian probabilistic framework is deployed to evaluate uncertainty by utilizing Markov Chain Monte Carlo sampling. A recursive Bayesian inference framework iteratively updates the statistical parameters of assumed lognormal distribution as data is processed in discrete batches. The example bridge that instrumented to carry out this study is located at Louisiana State University parking area. The results demonstrate the feasibility of using cost-effective technologies to estimate the dynamic parameters of a flexible structure while rigorously quantifying the associated uncertainties.

1 INTRODUCTION

Data acquisition systems are one of the necessary components of Structural Health Monitoring (SHM). SHM system delivers measurements that support the early detection of structural irregularities, thereby providing a robust empirical basis for informed maintenance planning and systematic risk management (An et al., 2019; Debees et al., 2024; Fujino et al., 2019). Despite their clear advantages, the widespread adoption of these systems remains limited, primarily

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due to the considerable costs involved in acquiring, installing, calibrating, and maintaining sensor networks, alongside the ongoing need for specialized technical personnel (Agdas et al., 2016; Fujino et al., 2019; Sun et al., 2017). As an alternative, modern smartphones, which incorporate substantial computational power, large memory capacity, built-in batteries, and a variety of built-in sensors, represent a promising and cost-effective platform for structural seismic monitoring, offering new opportunities for structural response measurement at significantly reduced initial and operational costs (Shrestha et al., 2020).

In the context of modal parameter identification, numerous studies investigated the feasibility of using smartphone accelerometers as a low-cost alternative technique to conventional approaches (Alavi and Buttlar, 2019; Castellanos-Toro et al., 2018; Ervasti et al., 2011; Feldbusch et al., 2017; Mourcou et al., 2015; Ozer and Feng, 2017). Although these studies have successfully demonstrated estimation of modal parameters with low-cost devices by applying deterministic methods, the uncertainty associated with these parameters has not been addressed. Since these devices have higher noise density, system identification may be influenced by the low signal-to-noise ratio characteristics of these devices, necessitating a different treatment in the identification process if compared to traditional systems (AlGadi et al., 2025; Dorvash and Pakzad, 2012). The objective of this paper is to determine the first four modes of a flexible pedestrian bridge and quantify the associated uncertainty with the data.

To achieve this, ten smartphones were deployed at equal spacing along the bridge deck with equal distance on the ground, and ambient excitation was generated through random pedestrian activities, including walking, running, and cycling. In this study, we aimed to identify whether it is feasible to deploy smartphones to estimate modal parameters of the pedestrian bridge under random ambient-induced excitation or not, and to compute the minimum measurement duration required for posterior mean and standard deviation parameters to converge, meaning that this amount of data is sufficient to identify associated uncertainties in the data.

2 METHODOLOGY

The methodological workflow (Figure 1) of this study consists of four main stages: (1) full-scale data acquisition under pedestrian-induced excitation; (2) preprocessing of vibration data; (3) modal parameter estimation using Frequency Domain Decomposition (FDD); and (4) Bayesian inference for uncertainty quantification.

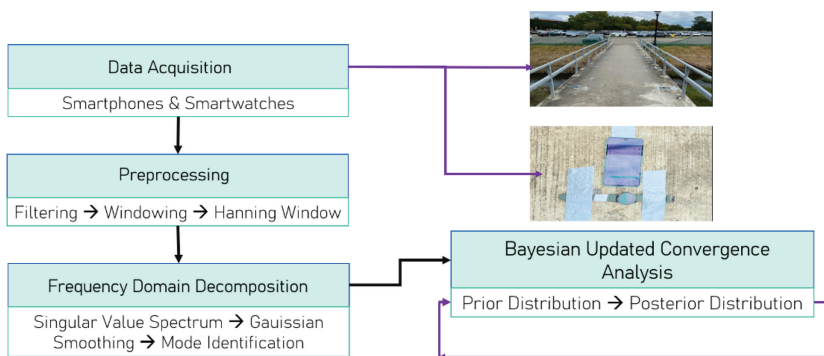


Figure 1. Methodological framework.

2.1 Data acquisition

Vibration data were measured by using smartphones that were fixed along the surface of the bridge. These devices were evenly placed across the bridge deck to capture vibration data

consistently along the span. About randomly induced pedestrian vibration, which was generated by a group of six people, consisting of a pair of runners, walkers, and a pair of riders.

Table 1. Details of the experimental test.

Excitation	Duration	Number of Devices	Sampling Rate	Phone (Samsung A36) IMU Sensor
2 Walker 2 Runner 2 Bicycle	300 s	10 smartphones	~62.25 Hz	TDK ICM-42632M

2.2 Preprocessing and Frequency Domain Decomposition (FDD)

Before the estimation of modal parameters with FDD, it is necessary to process the raw data, making it ready for FDD. First, the raw signals are detrended to remove a constant offset, eliminating low-frequency response that could obscure the structural response, ensuring the vibration signal is centered around zero. Following this, a Butterworth band-pass filter (0.5-30 Hz) was applied to remove both very low frequency components and high frequency components that do not contribute to the structural response. The trimmed signals were then divided into 60 windows that are restricted to 5-second intervals without overlapping (Luleci et al., 2023). A Hanning window was applied to each window to reduce spectral leakage, leading to sharp and distinct peaks at the natural frequencies. Finally, the Welch method is used for each window to build the spectral density matrix. Once the frequency values for all 60 windows were obtained, these frequencies were used as input data in discrete batches for Bayesian Updated Convergence analysis to quantify the uncertainty of the frequency distribution.

2.3 Bayesian probabilistic approach

To quantify the associated uncertainty for f_1, f_2, f_3 , and f_4 , the obtained frequency values from FDD were processed in non-overlapping batches of three windows; thus, each batch contains three frequency data points for each mode. Since frequency values can not be negative, our prior assumption for the Probability Density Function (PDF) of frequency values is a Lognormal Distribution. The main idea of this method is that “today’s posterior is tomorrow’s prior”. By utilizing Markov Chain Monte Carlo (MCMC) sampling, it iteratively updates the posterior estimates for the mean and standard deviation, allowing for dynamic characterization of the underlying data distribution.

The process begins by assuming that frequency data follows a lognormal distribution, and our probabilistic model tracks two parameters that are the average mean of log-transformed data and the standard deviation of the log-transformed data. A likelihood function is then used to evaluate the probability of each batch of observed sensor data relative to these parameter estimates. By mathematically combining these initial assumptions with the evidence from observed data, it is resulting in a generated posterior distribution which represents a more accurate and updated understanding of the modal parameters.

3 RESULTS

This section presents the experimental findings in two stages. Initially, the mode identification results obtained via FDD using the full 300-second datasets are compared with those recorded by conventional ultra-sensitive accelerometer devices, which serve as the benchmark in this study. The FDD results for both approaches are illustrated in Figures 2 and 3. In the second stage, a Bayesian statistical approach is applied to update and refine the modal identification results obtained from windowed FDD analysis. As observed in Figure 3, the smartphones were unable to capture the first shoulder peak at approximately 1.4 Hz, which is a weakly

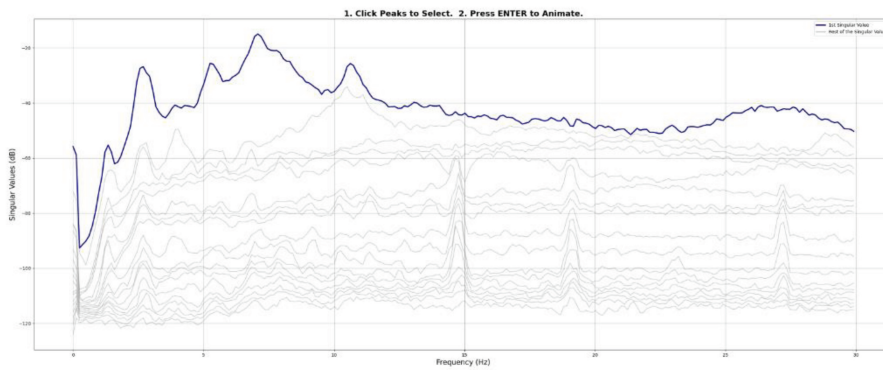


Figure 2. Singular value spectra based on conventional sensors.

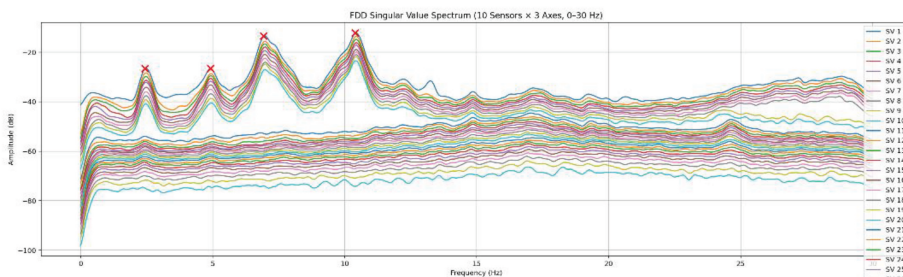


Figure 3. Singular value spectra based on smartphones.

excited peak. Identifying this can be attributed to the superior noise sensitivity of the conventional SHM sensors, which enables them to detect such low-amplitude peaks. However, the remaining singular value peaks are approximately same between both approaches and are therefore suitable for further statistical analysis.

Secondly, the FDD process was executed using 5-second non-overlapping windows, where four distinct modes of the pedestrian bridge were estimated across 60 separate windows (Figure 4). Once the frequency values are estimated, these values are processed with a Bayesian Probabilistic approach to quantify the associated uncertainty in the data. However, MCMC simulations failed to converge (Figure 5), preventing an accurate quantification of uncertainty. After investigating the FDD results, it is revealed that smartphones could not capture data accurately within the first 5-10 seconds, leading to an inaccurate frequency estimate in the first window that acted as a statistical outlier compared to the subsequent windows (Figure 6). Consequently, the duration of the first window was increased to 15 seconds in the second attempt, thereby accurate frequency values were successfully established in the first window and avoided any statistical outliers (Figure 7).

By observing Figure 6, anomalies in the results of the first window can be noticed easily because the obtained singular spectrum value in the first window is different than the rest of the windows; thus, MCMC was not able to quantify the uncertainty in the data and failed to converge (Figure 5). After changing the duration of the first window to 15 seconds, all the windows were reasonable, and the obtained frequency values were the characteristic modal parameters of the pedestrian bridge, meaning that those data can be used in the simulation part to quantify the associated uncertainties.



Figure 4. Singular spectrum values with 5 seconds windowing.

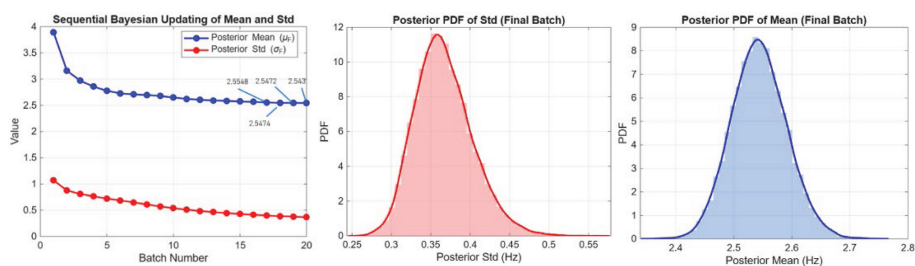


Figure 5. Marginal distributions and convergence plot of mean and standard deviation for mode 1 with 5 seconds windowing.

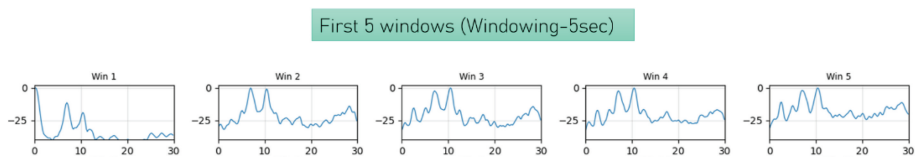


Figure 6. First five Singular Spectrum values with 5 second windowing.

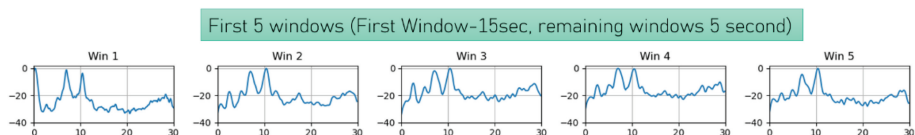


Figure 7. First 5 singular spectrum values with modified duration for the first window.

Figures 8-11 demonstrates the results of the Bayesian inference statistical analysis with modified windowing settings as described above.

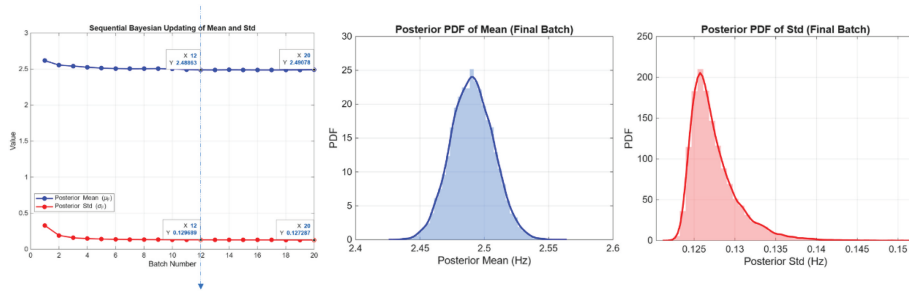


Figure 8. Convergence plot and marginal distributions of mean and standard deviation for mode 1.

Figure 8 provides insight into the marginal distributions of f_1 and demonstrates the duration required to obtain reliable results, illustrating where statistical parameters converge, and the associated uncertainty is quantified. For mode 1, the simulation converges efficiently after only 12 batches. Given that each batch incorporates data from three separate windows, a total recording duration of 190 seconds is enough to collect feasible data and accurately characterize the first mode of the pedestrian bridge.

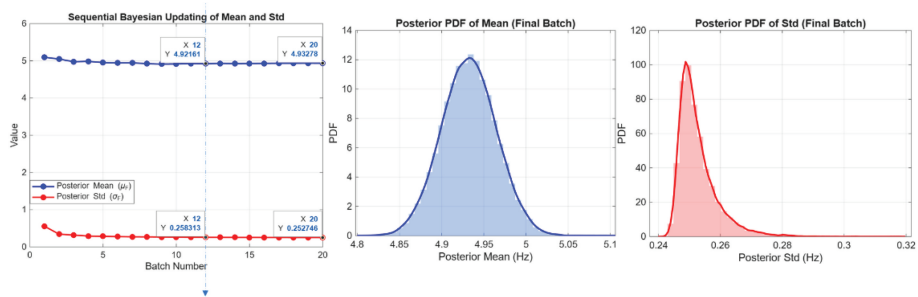


Figure 9. Convergence plot and marginal Distributions of mean and standard deviation for mode 2.

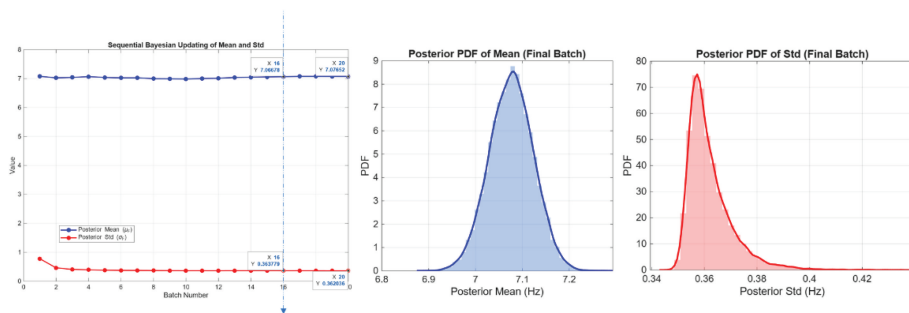


Figure 10. Convergence plot and marginal distributions of mean and standard deviation for mode 3.

Similarly, the remaining modes f_2 and f_4 exhibit the same pattern as f_1 , demonstrating that smartphone-integrated sensors can provide reliable data within 190 seconds to accurately estimate the true parameters of these modes. However, f_3 required a more extensive dataset than the previous modes; specifically, 16 batches (equivalent to 240 seconds of recording duration)

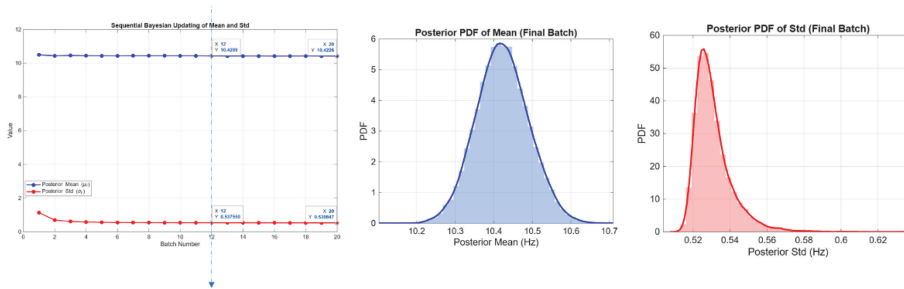


Figure 11. Convergence plot and marginal distribution of mean and standard deviation for mode 4.

were necessary to reach convergence for the mean and standard deviation of f_3 and to rigorously quantify the associated uncertainties. These results are summarized in Table 2.

Table 2. Results of Bayesian probabilistic analysis.

Modes	Number of Seconds required for Convergence	Posterior PDF of Mean	Posterior PDF of Standard Deviation
Mode 1	190	Normal	Lognormal
Mode 2	190	Normal	Lognormal
Mode 3	240	Normal	Lognormal
Mode 4	190	Normal	Lognormal

Regarding the marginal distributions of the four frequency values, it was assumed that the data would follow a lognormal distribution to reflect the physical constraint that frequency values are non-negative. Consequently, results indicate that the marginal distributions are consistent across f_1 , f_2 , f_3 , and f_4 . Specifically, the posterior PDF for the mean exhibits a normal distribution profile, while the posterior PDF for the standard deviation follows a lognormal distribution, highlighting the convergence behavior of the MCMC sampler in characterizing the uncertainty associated with each modal parameter.

The summarized results of the Bayesian probabilistic approach (Table 2) demonstrate that smartphones provide a reliable, cost-effective, and easily deployable alternative for the assessment of pedestrian bridges. Their accuracy in system identification and uncertainty quantification can be effectively employed to support timely maintenance and ensure long-term sustainability of these structures.

4 CONCLUSION

This paper explored the feasibility and accuracy of using low-cost integrated smartphones for system identification and quantification of associated uncertainties with the modal parameters of a case study. Although smartphones are not primarily designed to perform modal parameter estimation, observations showed that this device can capture data that is sufficiently accurate, statistically stable, and feasible enough to quantify uncertainty. Consequently, this approach offers a viable alternative to conventional modal parameter estimation over conventional techniques. To validate the framework, field experiments were conducted on a pedestrian bridge where vibrations were excited by random pedestrian activities (running, walking, bicycling) while smartphones were positioned evenly across the structure to record a 300-second duration of data. The main findings of this study are listed below:

- While not originally developed for condition assessment, low-cost integrated smartphones can provide sufficient accuracy and statistical stability for modal identification. The feasibility, cost-effectiveness, and ease of installation of smartphone-based monitoring support continuous maintenance, making it an alternative approach to traditional SHM techniques.
- With the application of a recursive Bayesian inference framework, supported by MCMC sampling, it was determined that a recording duration of 190 seconds is sufficient for the convergence of f_1 , f_2 , and f_4 , while f_3 required up to 240 seconds (16 batches) to reach statistical convergence to quantify associated uncertainty within the data. Additionally, observations exhibited that marginal distributions are consistent across all four modes.
- To ensure MCMC convergence and avoid possible errors originating from sensor stabilization periods (sensors don't capture accurate data), the first window duration increased from 5 to 15 seconds. This adjustment was critical in avoiding inaccurate early-stage readings.

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